

CMOS LOGIC GATE DESIGN & ANALYSIS

Cadence Virtuoso Simulation and Power-Propagation Delay Analysis

1. CMOS Inverter

The CMOS inverter is the reference circuit. It contains one PMOS pull-up transistor and one NMOS pull-down transistor. The common gate node receives the input and the common drain node provides the output.

Figure 1.1 – CMOS_Inverter_Schematic

The Cadence Virtuoso schematic of the CMOS inverter.

Explanation: The PMOS is connected to the supply and the NMOS to ground. The input drives both transistor gates. When the input is LOW, the PMOS conducts and the output becomes HIGH. When the input is HIGH, the NMOS conducts and the output becomes LOW.

Figure 1.2 – Inverter Functional Transient Waveform

Input and output voltage waveforms plotted against time.

Explanation: The output is complementary to the input. At approximately 1.8 V input, the output is approximately 0 V; at approximately 0 V input, the output returns to approximately 1.8 V. This verifies NOT-gate functionality.

Figure 1.3 – Inverter Transient Voltage and Total Power Waveform

The logic waveforms together with the transient power waveform.

Explanation: This picture is used to observe power behavior during switching. Dynamic power appears mainly when the circuit changes state.

Figure 1.4 – Inverter Average-Power Result

Cadence's calculated average power value.

Explanation: The reported average power is 424.1E-9 W, equivalent to 424.1 nW.

Figure 1.5 – Inverter tPLH Measurement

Marker-based measurement for input falling and output rising.

Explanation: The two displayed marker times are 15.07702 ms and 15.07907 ms. Their difference is 2.05 μ s, reported as tPLH.

Figure 1.6 – Inverter tPHL Measurement

Marker-based measurement for input rising and output falling.

Explanation: The two displayed marker times are 10.02088 ms and 10.02213 ms. Their difference is 1.25 μ s, reported as tP

2. 2-Input CMOS NAND Gate

The 2-input NAND gate uses two PMOS transistors in parallel in the pull-up network and two NMOS transistors in series in the pull-down network. The output becomes LOW only when both A and B are HIGH.

Figure 2.1 – NAND Transistor-Level Schematic

The Cadence schematic of the 2-input NAND gate.

Explanation: The parallel PMOS network provides a pull-up path when either input is LOW. The series NMOS path provides a complete pull-down path only when both inputs are HIGH.

Figure 2.2 – NAND Functional Transient Waveform

What the picture shows: A, B and Y transient waveforms.

Explanation: The output remains HIGH for input combinations other than A = B = HIGH and becomes LOW when both inputs are HIGH. This verifies NAND functionality.

Figure 2.3 – NAND Power/Logic Transient Waveform

Logic signals and total transient power.

Explanation: The plot allows switching activity and corresponding power behavior to be observed together.

Figure 2.4 – NAND Average-Power Result

Cadence average-power calculation.

Explanation: The reported average power is 227.7E-9 W, equivalent to 227.7 nW.

Figure 2.5 – NAND Delay Measurement 1

One switching-direction marker measurement

Explanation: The displayed marker times are 3.5549898 ms and 3.5550170 ms. Their difference is 0.0272 μ s = 27.2 ns.

Figure 2.6 – NAND Delay Measurement 2

The opposite switching-direction marker measurement.

Explanation: The displayed marker times are 14.01498 ms and 14.01825 ms. Their difference is 3.27 μ s.

3. 2-Input CMOS NOR Gate

The 2-input NOR gate uses two PMOS transistors in series and two NMOS transistors in parallel. The output is HIGH only when both inputs are LOW.

Figure 3.1 – NOR Transistor-Level Schematic

The Cadence schematic of the 2-input NOR gate.

Explanation: The series PMOS network requires both inputs to be LOW to establish a complete pull-up path. The parallel NMOS network pulls the output LOW if either input is HIGH.

Figure 3.2 – NOR Functional Transient Waveform

A, B and Y transient waveforms.

Explanation: Any HIGH input drives the output LOW, while $A = 0$ and $B = 0$ produces a HIGH output. This verifies NOR functionality.

Figure 3.3 – NOR Power Transient Waveform

Logic and total transient power waveforms.

Explanation: The waveform is used to examine switching power behavior during the transient simulation.

Figure 3.4 – NOR Average-Power Result

Cadence average-power calculation.

Explanation: The reported average power is $283.8\text{E-}9$ W, equivalent to 283.8 nW.

Figure 3.5 – NOR tPHL Measurement

Marker measurement for input rising and output falling.

Explanation: The displayed marker times are 12.537702 ms and 12.534854 ms. Their absolute difference is 2.848 μs .

Figure 3.6 – NOR tPLH Measurement

Marker measurement for input falling and output rising.

Explanation: The displayed marker times are 5.012418 ms and 5.015139 ms. Their difference is 2.721 μs .

4. Power Analysis

The Cadence average-power results are compared below. These values are specific to the supplied simulation conditions, transistor models, dimensions, input activity and loading.

Power ranking: NAND < NOR < Inverter. The NAND gate has the lowest measured average power, followed by NOR, while the inverter has the highest measured average power.

5. Propagation Delay Analysis

Propagation delay is the time difference between corresponding input and output transition reference points. Because rising and falling transitions can differ, tPHL and tPLH are retained separately.

Delay Calculations

Inverter: $t_{PHL} = |10.02213 - 10.02088| = 1.25 \mu\text{s}$; $t_{PLH} = |15.07907 - 15.07702| = 2.05 \mu\text{s}$; Mean = $1.65 \mu\text{s} = 1650 \text{ ns}$.

NAND: $t_{PHL} = |14.01825 - 14.01498| = 3.27 \mu\text{s}$; $t_{PLH} = |3.5550170 - 3.5549898| = 0.0272 \mu\text{s} = 27.2 \text{ ns}$; Mean = $1.6486 \mu\text{s}$.

NOR: $t_{PHL} = |12.537702 - 12.534854| = 2.848 \mu\text{s}$; $t_{PLH} = |5.015139 - 5.012418| = 2.721 \mu\text{s}$; Mean = $2.7845 \mu\text{s}$.

6. Overall Performance Comparison

The NAND delay measurements are strongly asymmetric, so both individual values should be considered when interpreting performance.

7. Power–Delay Trade-Off Analysis

Final observation: No single gate provides the lowest value for both measured power and delay. The inverter favors speed with the lowest mean delay in the supplied report, whereas the NAND gate favors low power. The NOR gate shows the highest mean displayed delay. Therefore, gate selection depends on whether the application prioritizes switching speed, low power consumption, or balanced behavior.

8. Final Conclusion

The Cadence Virtuoso evidence demonstrates successful transistor-level implementation and transient simulation of a CMOS inverter, 2-input NAND gate and 2-input NOR gate. The functional waveforms verify the expected Boolean operations. Cadence average-power measurements provide a direct comparison of power consumption, while marker-based measurements provide tPHL and tPLH values. The results demonstrate the practical CMOS design trade-off between circuit complexity, switching speed and power dissipation. For the supplied simulation set, the NAND gate has the lowest measured average power, while the inverter has the lowest mean displayed delay as stated in the project report.

Key Simulation Results

Gate	Average Power	tPHL	tPLH	Mean Displayed Delay
CMOS Inverter	424.1 nW	1.25 μ s	2.05 μ s	1.65 μ s
2-input NAND	227.7 nW	3.27 μ s	0.0272 μ s (27.2 ns)	1.6486 μ s
2-input NOR	283.8 nW	2.848 μ s	2.721 μ s	2.7845 μ s

Power ranking: NAND < NOR < Inverter. The NAND gate has the lowest measured average power, followed by NOR, while the inverter has the highest measured average power.

The supplied report describes the inverter as the speed-oriented result, while the NAND gate provides the lowest measured power. The NOR gate has the highest mean displayed delay in this dataset.